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import java.io.*;
import java.util.ArrayList;
import java.util.InputMismatchException;
import java.util.List;
import java.util.Scanner;

public class Task5 implements Serializable {

    public class student implements Serializable {
        private String name;
        private int rollNo;
        private String grade;

        public student(String name, int rollNo, String grade) {
            this.name = name;
            this.rollNo = rollNo;
            this.grade = grade;
        }

        public String getName() {
            return name;
        }

        public void setName(String name) {
            this.name = name;
        }

        public int getRollNo() {
            return rollNo;
        }

        public void setRollNo(int rollNo) {
            this.rollNo = rollNo;
        }

        public String getGrade() {
            return grade;
        }

        public void setGrade(String grade) {
            this.grade = grade;
        }

        @Override
        public String toString() {
            return "student" +
                "{" +
                "name='" + name + '\'' +
                ", rollNo=" + rollNo +
                ", grade='" + grade + '\'' +
                '}';
        }
    }

    public class studentManagementSystem {
        private List<student> students;
        private final String filePath = "students.ser";

        public studentManagementSystem() {
            this.students = new ArrayList<>();
            load();
        }

        public void add(student pupil) {
            students.add(pupil);
            save();
        }
    }
}

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    public void remove(int rollNo) {
        students.removeIf(student -> student.getRollNo() == rollNo);
        save();
    }

    public student search(int rollNo) {
        for (student pupil : students) {
            if (pupil.getRollNo() == rollNo) {
                return pupil;
            }
        }
        return null;
    }

    public void displayAll() {
        for (student pupil : students) {
            System.out.println(pupil);
        }
    }

    private void save() {
        try (ObjectOutputStream oos = new ObjectOutputStream(new
FileOutputStream(filePath))) {
            oos.writeObject(students);
        } catch (IOException e) {
            System.out.println("error in saving: " + e.getMessage());
        }
    }

    private void load() {
        try (ObjectInputStream ois = new ObjectInputStream(new FileInputStream(filePath)))
{
            students = (List<student>) ois.readObject();
        } catch (IOException | ClassNotFoundException e) {
            System.out.println("no data found");
        }
    }
}

public static void main(String[] args) {
    Task5 task5 = new Task5();
    studentManagementSystem x = task5.new studentManagementSystem();
    Scanner reader = new Scanner(System.in);

    while (true) {
        System.out.println("student management system");
        System.out.println("1: add a new student");
        System.out.println("2: remove a student");
        System.out.println("3: search for a student");
        System.out.println("4: display all students");
        System.out.println("5: exit");
        System.out.print("choose an option: ");

        int choice;
        try {
            choice = reader.nextInt();
        } catch (InputMismatchException e) {
            System.out.println("invalid, please enter a number: ");
            reader.nextLine();
            continue;
        }

        reader.nextLine();

        switch (choice) {
            case 1:

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        System.out.print("enter name: ");
        String name = reader.nextLine();
        if (name.isEmpty()) {
            System.out.println("name cannot be empty");
            break;
        }

        System.out.print("enter roll number: ");
        int rollNo;
        try {
            rollNo = reader.nextInt();
        } catch (InputMismatchException e) {
            System.out.println("invalid, please enter a valid roll number: ");
            reader.nextLine();
            break;
        }

        reader.nextLine();

        System.out.print("enter grade: ");
        String grade = reader.nextLine();
        if (grade.isEmpty()) {
            System.out.println("grade cannot be empty");
            break;
        }

        student newPupil = task5.new student(name, rollNo, grade);
        x.add(newPupil);
        System.out.println("student added");
        break;
    case 2:
        System.out.print("enter roll number to remove: ");
        int removeRollNo;
        try {
            removeRollNo = reader.nextInt();
        } catch (InputMismatchException e) {
            System.out.println("invalid, please enter a valid roll number: ");
            reader.nextLine();
            break;
        }

        x.remove(removeRollNo);
        System.out.println("student removed");
        break;
    case 3:
        System.out.print("enter roll number for search: ");
        int searchRollNo;
        try {
            searchRollNo = reader.nextInt();
        } catch (InputMismatchException e) {
            System.out.println("invalid, please enter a valid roll number: ");
            reader.nextLine();
            break;
        }

        student found = x.search(searchRollNo);
        if (found != null) {
            System.out.println("student found: " + found);
        } else {
            System.out.println("not found");
        }
        break;
    case 4:
        x.displayAll();
        break;
    case 5:
        System.out.println("exiting");

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        System.exit(0);  
    default:  
        System.out.println("invalid, try again");
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    }
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  }
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}
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}
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